

MICHAEL (mike) NEUDER

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🌐 Princeton, New Jersey, USA

EDUCATION

Princeton University <i>Doctoral student, Computer Science</i> ↳ Under the supervision of Prof. Matt Weinberg ↳ Supported by an Ethereum Foundation PhD Fellowship (FY25-2276)	Aug. 2025 - Present Princeton, NJ
Harvard University <i>Master of Science, Computational Science</i>	Aug. 2020 - May. 2021 Cambridge, MA
University of Colorado <i>Bachelor of Arts, Computer Science</i> <i>Bachelor of Arts, Mathematics</i>	Aug. 2015 - May. 2020 Boulder, CO

ACADEMIC EXPERIENCE

EconCS Group, Harvard University <i>Research Assistant supervised by Prof. David C. Parkes</i>	Aug. 2019 - Aug. 2021 Cambridge, MA
Santa Fe Institute <i>Undergraduate Research Fellow: Summer 2018 REU</i>	Jun. 2018 - Aug. 2018 Santa Fe, NM
Bradley Lab, University of Colorado <i>Research Assistant supervised by Prof. Elizabeth Bradley</i>	Apr. 2017 - Aug. 2021 Boulder, CO
Mozer Lab, University of Colorado <i>Research Assistant supervised by Prof. Michael Mozer</i>	Mar. 2017 - May. 2019 Boulder, CO

INDUSTRY EXPERIENCE

The Ethereum Foundation <i>Researcher – Applied Research Group (ARG)</i>	Mar. 2023 - May. 2025 New York, NY
Google <i>Software Engineer (L3 → L4) – Cloud Storage</i> <i>Software Engineering Intern – Network Infrastructure</i> <i>Software Engineering Intern – Flights</i> <i>Software Engineering Intern – Mobile Device Management</i>	Aug. 2021 - Feb. 2023 (Cambridge, MA) Summer 2020 (remote) Summer 2019 (Cambridge, MA) Fall 2018 (Sunnyvale, CA)
Lockheed Martin & LASP¹ <i>Software Engineering Intern</i>	Feb. 2017 - Oct. 2017 Boulder, CO

JOURNAL AND CONFERENCE PUBLICATIONS (REVERSE CHRONOLOGICAL)

- P. Garimidi, **M. Neuder**, T. Roughgarden, “Beyond Winner-Take-All Procurement Auctions,” (2026) *Financial Cryptography and Data Security (FC’26)*. <https://arxiv.org/pdf/2603.27779>.
- M. Bahrani, **M. Neuder**, S. M. Weinberg, “Selfish mining under general stochastic rewards,” (2025) *Advances in Financial Technologies (AFT’25)*.
<https://doi.org/10.4230/LIPIcs.AFT.2025.20>. (full version on arxiv: <https://arxiv.org/abs/2502.20360>.)

¹Laboratory of Atmospheric and Space Physics: <https://lasp.colorado.edu/home/>.

- **M. Neuder**, M. Pai, and M. Resnick, “Optimizing Exit Queues for Proof-of-Stake Blockchains: A Mechanism Design Approach,” (2024) *Advances in Financial Technologies (AFT’24)*.
<https://doi.org/10.4230/LIPIcs.AFT.2024.20>.
- Z. Fan, F. Marmolejo-Cossio, D. J. Moroz, **M. Neuder**, R. Rao, and D. C. Parkes, “Strategic Liquidity Provision in Uniswap v3,” (2023) *Advances in Financial Technologies (AFT’23)*.
<https://doi.org/10.4230/LIPIcs.AFT.2023.25>.
- **M. Neuder**, E. Bradley, E. Dlugokencky, J. W. C. White, J. Garland, “Detection of Local Mixing in Time Series using Permutation Entropy,” (2021) *Physical Review E* 103.
<https://doi.org/10.1103/PhysRevE.103.022217>.
- **M. Neuder**, D. J. Moroz, R. Rao, D. C. Parkes, “Defending Against Malicious Reorgs in Tezos Proof-of-Stake,” (2020) *ACM Conference on Advances in Financial Technologies (AFT’20)*.
<https://doi.org/10.1145/3419614.3423265>.
- **M. Neuder**, D. J. Moroz, R. Rao, D. C. Parkes, “Selfish Behavior in the Tezos Proof-of-Stake Protocol,” (2020) *Cryptoeconomic Systems (CES’20) Conference*. <https://arxiv.org/pdf/1912.02954>.
- J. Garland, T. Jones, **M. Neuder**, J. W. C. White, E. Bradley, “An information-theoretic approach to extracting climate signals from deep polar ice cores,” (2019) *Chaos: An Interdisciplinary Journal of Nonlinear Science* 29:101105. <https://doi.org/10.1063/1.5127211>.
- J. Garland, T. Jones, **M. Neuder**, V. Morris, J. W. C. White, E. Bradley, “Anomaly Detection in Paleoclimate Records using Information Theory,” (2018) *Entropy* 20(12):931. <https://doi.org/10.3390/e20120931>.

WORKSHOP PAPERS, LETTERS, AND PREPRINTS

- M. Bahrani, **M. Neuder**, S. M. Weinberg, “Adversarial procurement in blockchains,” (2026) *preprint*.
<https://arxiv.org/pdf/2605.05559>.
- P. Garimidi, **M. Neuder**, T. Roughgarden, “Tullock Contests in the Wild: Applications in Blockchains,” (2025) *ACM SIGecom Exchanges, Vol. 23, No. 1*.
https://www.sigecom.org/exchanges/volume_23/1/GARIMIDI.pdf.
- **M. Neuder**, D. J. Moroz, R. Rao, D. C. Parkes, “Low-cost attacks on Ethereum 2.0 by sub-1/3 stakeholders,” (2020) *Workshop on Game Theory in Blockchain, Conference on Web and Internet Economics (WINE)*.
<https://arxiv.org/pdf/2102.02247>.

ACADEMIC TALKS

- **Beyond Winner-Take-All Procurement Auctions**
↳ Financial Cryptography and Data Security, *FC 2026*. [slides](#).
- **Selfish mining under general stochastic rewards**
↳ Advances in Financial Technology, *AFT 2025*. [recording](#). [slides](#).
- **Optimizing Exit Queues for Proof-of-Stake Blockchains**
↳ Advances in Financial Technology, *AFT 2024*. [slides](#).
- **Low-cost attacks on Ethereum 2.0 by sub-1/3 stakeholders**
↳ Workshop on Game Theory in Blockchain, *WINE 2020*. [recording](#). [slides](#).
- **Defending Against Malicious Reorgs in Tezos**
↳ ACM Advances in Financial Technology, *AFT 2020*. [recording](#). [slides](#).
- **Selfish Behavior in the Tezos PoS Protocol**
↳ Cryptoeconomic Systems Conference, *CES 2020*. [recording](#). [slides](#).
- **Animal Tracking using Deep Learning**
↳ Santa Fe Institute REU Presentation, *SFI 2018*. [recording](#).

AWARDS & FUNDING

2025-2030 Ethereum Foundation PhD Fellowship (award FY25-2276; \$200,000).

2020 Computer Science Discovery Learning Award (University of Colorado).
2019 Sieglinde Talbott Haller Scholarship in Mathematics (University of Colorado).
2019 Honorable Mention: Computing Research Assoc. Outstanding Undergraduate Researcher. [link.](#)
2017 Phi Beta Kappa (University of Colorado). [link.](#)
2015-2020 President Joseph A. Sewall Esteemed Scholar Award (University of Colorado). [link.](#)
2015-2020 Dean's List (University of Colorado). [link.](#)

PROFESSIONAL SERVICE

Columbia CryptoEconomics Workshop (CCE). ↳ 2023, 2024, 2025	Organizing Committee. link.
ACM Conference on Economics and Computation (EC). ↳ 2026	Program Committee. link.
Designing Defi (D^2) Conference. ↳ 2026	Program Committee. link.
The Latest in DeFi Research (TLDR) Conference. ↳ 2024, 2025	Program Committee. link.
Crypto & Blockchain Economics Research Forum (CBER) Conference. ↳ 2025, 2026	Program Committee. link.
Workshop on Decentralized Finance (DeFi) at Finanacial Cryptography. ↳ 2026	Program Committee. link.
Princeton Blockchain Special Interest Group (SIG). ↳ AY 2025-2026	Organizer.
Symposium On Discrete Algorithms (SODA). ↳ 2026	External Reviewer.
ACM Symposium on Principles of Distributed Computing (PODC). ↳ 2026	External Reviewer.
Crypto Academic Summer School at Edge City. ↳ 2024	Organizing Committee. link.
dYdX MEV Committee. ↳ 2023	Committee Member. link.

BLOCKCHAIN ARTICLES (NON-PEER REVIEWED) (BY TOPIC)

Proof-of-Stake

ETH, the asset

- My (e)thesis: settlement, data availability, execution – in that order. [link.](#) [tweet.](#)
- Issuance Issues – Tertiary Treatise. [link.](#) [tweet.](#)
- Issuance Issues – Subsequent Soliloquy. [link.](#) [tweet.](#)
- Issuance Issues – Initial Issue. [link.](#) [tweet.](#)

Ethereum Consensus

- Concurrent Block Proposers in Ethereum, *with Max Resnick.* [link.](#) [tweet.](#)
- Rollup-Centric Roadmap (2024 version), *with Alex Stokes.* [link.](#) [tweet.](#)
- A set-theoretic view of Ethereum coteries. [link.](#) [tweet.](#)
- Time, slots, and the ordering of events in Ethereum, *with Georgios Konstantopoulos.* [link.](#) [tweet.](#)

Increase the MAX_EFFECTIVE_BALANCE

- EIP-7251: Increase the MAX_EFFECTIVE_BALANCE. [link.](#)
- A modest proposal, *with Francesco D'Amato, Aditya Asgaonkar, Justin Drake.* [link.](#) [tweet.](#)
- Slashing penalty analysis; EIP-7251, *with Barnabe Monot.* [link.](#) [tweet.](#)
- Validator consolidation in EIP-7251, *with Francesco D'Amato, Mikhail Kalinin.* [link.](#) [tweet.](#)
- FAQ on EIP-7251, *with Francesco D'Amato, Mikhail Kalinin, dAppLion.* [link.](#) [tweet.](#)

Exit/Withdrawal queues

- EIP-7922: Dynamic exit queue rate limit, *with Mikhail, Mallesh.* [link.](#)
- Adding flexibility to Ethereum's exit queue, *with Mikhail Kalinin, Mallesh Pai.* [link.](#) [tweet.](#)

- ELI5: Ethereum Validator Exits, *with Mallesh Pai*. [link](#). [tweet](#).
- Liquid staking & restaking
 - The Risks of LRTs, *with Tarun Chitra*. [link](#). [tweet](#).
 - Musings on “two-tiered” staking, a native Liquid Staking Token design. [link](#). [tweet](#).
 - Magnitude and direction of Lido attack vectors. [link](#). [tweet](#).
- Data availability/blobs
 - Blob gossip and validation before and after PeerDAS. [link](#). [tweet](#).
 - On the future of the blob mempool, *with Julian Ma*. [link](#). [tweet](#).
- Maximal Extractable Value (MEV)**
- Enshrined Proposer-Builder Separation (ePBS)
 - Why enshrine Proposer-Builder Separation, *with Justin Drake*. [link](#). [tweet](#).
 - Payload-timeliness committee (PTC), *with Francesco D’Amato*. [link](#). [tweet](#).
 - Equivocation attacks in mev-boost and ePBS, *with Francesco D’Amato*. [link](#). [tweet](#).
 - Relays in a post-ePBS world, *with Jon, Hasu, Tomasz, Chris, Toni*. [link](#). [tweet](#).
 - ePBS – the infinite buffet. [link](#). [tweet](#).
 - Consider the ePBS. [link](#). [tweet](#).
 - Decoupling ExecutionPayload validity from Data Availability. [link](#). [tweet](#).
- Censorship resistance
 - EIP-7547: Inclusion lists. [link](#).
 - No free lunch – a new inclusion list design, *with Vitalik Buterin*. [link](#). [tweet](#).
 - Unconditional inclusion lists, *with Toni Wahrstatter*. [link](#). [tweet](#).
 - Resistance is not futile; CR in mev-boost. [link](#). [tweet](#).
 - Inclusion lists: execution, consensus, & engine spec overview. [link](#). [tweet](#).
 - Inclusion Lists PoC Specification. [link](#). [tweet](#).
 - The Case for ILECTRA. [link](#). [tweet](#).
- Relays
 - Optimistic Relay Proposal, *with Justin Drake*. [link](#).
 - An optimistic weekend. [link](#). [tweet](#).
 - Towards enshrined PBS – an optimistic roadmap. [link](#).
 - Optimistic relays and where to find them, *with Ankit Chiplunkar*. [link](#). [tweet](#).
 - Bid cancellations considered harmful, *with Thomas Thiery*. [link](#). [tweet](#).
- Procurement**
 - Capacity Oracles, *with Maryam Bahrani*. [link](#). [tweet](#).
 - On Ethereum Prover Market Design, *with Maryam Bahrani*. [link](#). [tweet](#).
 - On incentivizing anonymous participation, *with Maryam Bahrani*. [link](#). [tweet](#).
- Enshrined auction mechanisms**
 - Mechan-stein (alt. Franken-ism). [link](#). [tweet](#).
 - On block-space distribution mechanisms, *with Pranav Garimidi, Tim Roughgarden*. [link](#). [tweet](#).
 - Execution Tickets, *with Justin Drake*. [link](#). [tweet](#).
 - How I learned to stop worrying and love mev-burn, *with Justin Drake, Toni Wahrstatter*. [link](#). [tweet](#).
- Miscellanea**
 - Application-controlled execution: A case study on cancel prioritization, *with Maryam Bahrani*. [link](#). [tweet](#).
 - Overnight reverse repurchases and the Fed’s balance sheet. [link](#). [tweet](#).
 - Timing Games: Implications and Possible Mitigations, *with Caspar Schwarz-Schilling*. [link](#). [tweet](#).

BLOCKCHAIN PRESENTATIONS, PODCASTS, PANELS (REVERSE CHRONOLOGICAL)

Presentations

- How to get untrusted people to work for you – *Princeton DeCenter Spring Conference (April 2026)*. [slides](#).
- Blockchain governance in practice – *EGR 373 (Princeton) Guest talk. March 2026*. [slides](#).
- Selfish Mining under General Stochastic Rewards – *IC3 Winter Retreat. January 2026*. [slides](#).

◦ Adversarial procurement – <i>Columbia CryptoEconomics (CCE). December 2025.</i>	recording. slides.
◦ First Principles of Blockchains – <i>World Bank Session. November 2025.</i>	slides.
<u>EthCC[8] (July 2025) – Cannes, France</u>	
↳ Blobs and the future of the blob mempool – <i>EthCC main event.</i>	recording. slides.
↳ Blob futures and the blob mempool – <i>Blockspace Futures Day.</i>	recording. slides.
◦ Revisiting Ethereum’s Exit Queue – <i>Beam Chain Call #4. April 2025.</i>	recording. slides.
◦ Validator incentives – <i>Oxford-Harvard Conference on DeFi. January 2025.</i>	slides.
<u>DEVCON 7 (November 2024) – Bangkok, Thailand</u>	
↳ ETH is permissionless money – <i>DEVCON Main stage.</i>	recording. slides.
↳ The ticker is ETH – <i>Bankless Summit.</i>	recording. slides.
◦ Neutrality in Ethereum – <i>Harvard Club of New York City. October 2024.</i>	slides.
◦ Separating MEV and Staking Rewards – <i>CBER Webinar. September 2024.</i>	recording. slides.
◦ More slides about Ethereum Blockspace – <i>Scroll Protocol Symposium. August 2024.</i>	recording. slides.
◦ More slides about PBS – <i>MEV Workshop, Stanford Blockchain Conf. August 2024.</i>	recording. slides.
◦ Postmodern Staking – <i>9th Annual IC3 Blockchain Camp. June 2024.</i>	slides.
◦ Is restaking the new staking? – <i>CBER Forum Annual Conference. May 2024.</i>	recording. slides.
◦ State of the Ethereum Union (1/N perspective) – <i>ETHBoston. April 2024.</i>	slides.
<u>ETHDenver (March 2024) – Denver, USA</u>	
↳ Execution Tickets – <i>Beyond the Block (Titan Builder & Hashkey Capital).</i>	recording. slides.
↳ The Risks of LRTs – <i>Research Day (SevenX Ventures).</i>	recording. slides.
◦ No Free Lunch – <i>Columbia CryptoEconomics Working Session. December 2023.</i>	recording. slides.
◦ Enshrining PBS – <i>Center for Digital Finance & Technologies. December 2023.</i>	slides.
<u>DevConnect (November 2023) – Istanbul, Turkey</u>	
↳ Increase the MAX_EFFECTIVE_BALANCE – <i>EthStaker Staking Gathering.</i>	recording. slides.
↳ A set theoretic view of Ethereum coteries – <i>LidoConnect.</i>	recording. slides.
◦ Why it’s hard to enshrine PBS – <i>Archetype MEV Lunch. November 2023.</i>	recording. slides.
<u>EthCC[6] (July 2023) – Paris, France</u>	
↳ Towards Enshrined Proposer-Builder Separation – <i>EthCC main event.</i>	recording. slides.
↳ Ethereum PBS R&D Roadmap – <i>Modular Summit.</i>	recording. slides.
↳ Increase the MAX_EFFECTIVE_BALANCE – <i>Kiln Rendez-vous.</i>	recording. slides.
◦ Reorgs in PoS – <i>MEV Roast; Reorg Edition. August 2021.</i>	recording. slides.
Podcasts	
◦ Pectra Explained – <i>Coinbase Institutional Webinar. May 2025.</i>	recording.
◦ ETH is Permissionless Money – <i>Cryptoria. December 2024.</i>	English. Chinese.
◦ Ethereum’s North Star – <i>The Gwart Show. October 2024.</i>	recording.
◦ The Future of Ethereum: Is This The Right Track? – <i>Bankless. October 2024.</i>	recording.
◦ The Ethereum Roadmap is NOT Off Track! – <i>Bankless. September 2024.</i>	recording.
◦ ETH Insights: Discussing MaxEB – <i>Coinbase Webinar. April 2024.</i>	recording. article.
◦ PeepAnEIP: EIP-7547 – <i>Ethereum Cat Herders. April 2024.</i>	recording.
◦ Endgame 2.0: A Guide to Vitalik’s Ethereum Roadmap – <i>Bankless. February 2024.</i>	recording.
◦ We’re Pretty Sure Mike and Max Can Fix MEV – <i>The Gwart Show. February 2024.</i>	recording.
◦ PeepAnEIP: EIP-7251 – <i>Ethereum Cat Herders. February 2024.</i>	recording.
◦ Eigenlayer In 2024 (co-host) – <i>Bankless. December 2023.</i>	recording.
◦ An Incomplete Guide to PBS – <i>Uncommon Core 2. September 2023.</i>	recording.
Panels	
◦ Decentralization and pragmatism – <i>Princeton DeCenter Spring Conference. April 2026.</i>	
◦ Restaking Panel – <i>Staking Rewards: Staking Summit. November 2024.</i>	recording.
◦ Decentralization Panel – <i>Espresso: Beyond the Baselayer. March 2024.</i>	recording.

- Decentralize or Bust – *bloXroute & ETHStaker*. March 2024.
- Alignment Panel – *EigenLayer Restaking Summit*. November 2023.
- PBS & Beyond – *MEVDay Paris*. July 2023.

[recording.](#)
[recording.](#)
[recording.](#)

OPEN SOURCE CONTRIBUTIONS

EIP-7547: Inclusion Lists (python) ([repo](#))

- EIP (Ethereum Improvement Proposal).
- Spec Overview (Consensus, Execution, and Engine API specifications).
- Proof-of-Concept Specification.
- Compilation of Related Work.

[link.](#)
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EIP-7251: Increase the MAX_EFFECTIVE_BALANCE (python) ([repo](#))

- EIP (Ethereum Improvement Proposal).
- Minimal Spec Change (Consensus specifications).
- Compilation of Related Work.

[link.](#)
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MEV-boost Relay: Optimistic Processing (golang) ([repo](#))

- Main pull request & design documentation.
- Header-only parsing optimization.
- Pull request list.
- Proposal.
- Builder Onboarding.

[link.](#)
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Erigon (Ethereum Execution Layer Client) (golang) ([repo](#))

- Refactored req/resp domain network encoding.
- Consensus spec implementation.
- Pull request list.

[link.](#)
[link.](#)
[link.](#)

Prysm (Ethereum Consensus Layer Client) (golang) ([repo](#))

- Keymanager code-health refactor.
- Pull request list.

[link.](#)
[link.](#)

STRUMPACK (High-performance matrix math package) (C++) ([repo](#))

- Mixed-precision (float64 & float32) iterative refinement.
- Pull request list.
- Write-up.

[link.](#)
[link.](#)
[link.](#)

OptRBC (Optimal solutions in 2D Rayleigh-Benard Convection) (FORTRAN) ([repo](#))

- Multiprocessor implementation using [openmp](#).
- Write-up.

[link.](#)
[link.](#)

Image quality convolutional neural networks. (python) ([repo](#))

- Write-up.
- Example network.

[link.](#)
[link.](#)

MISCELLANEA: FUN WRITING & SOFTWARE

Writing

- Tattoo probabilistic analysis.
- Solo-staking rig.
- NBA draft probabilistic analysis.
- Shakespeare Zipf-ian analysis.

[link.](#) [tweet.](#)
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Software

- Monte Carlo numerical demonstrations. (python)
- Connect Four PyQT application. (python)
- Terminal implementation of games. (perl)
- Rubik's cube solver. (C++)

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